

A Negative Literature Answer to the General Isotropic Polynomial L^2 Question

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August 9, 2026

Status

`literature_already_answered` (full negative answer to the unrestricted general-isotropic version).

Original question

On page 4, immediately after Theorem 2 of Glazer–Mikulincer, *Anti-concentration of polynomials: dimension-free covariance bounds and decay of Fourier coefficients*, arXiv:2108.04268, the authors ask whether the same conclusion holds for general isotropic log-concave measures, perhaps with suitable symmetries [1]. The conclusion in question is a dimension-free estimate

$$\int_{\mathbb{R}^n} f^2 d\mu \geq C_d \text{coeff}_d(f)^2$$

for every degree- d homogeneous polynomial.

Decisive later counterexample

Page 38, Section 5.1 and Proposition 5.1 of the same authors' later paper, *Anti-concentration of polynomials: L^p balls and symmetric measures*, arXiv:2603.22664, explicitly show that lack of symmetry produces pathological polynomials [2]. Let ν_n be the uniform probability measure on the isotropic Euclidean ball in $\mathbb{R}^n$, put $\mu_n = \nu_n \otimes \nu_n$, and define on $\mathbb{R}^{2n}$

$$f_n(x) = \frac{1}{\sqrt{2n}} \left(\sum_{i=1}^n x_i^2 - \sum_{i=n+1}^{2n} x_i^2 \right).$$

The product measure μ_n is isotropic and log-concave, while f_n is homogeneous quadratic and

$$\text{coeff}_2(f_n)^2 = 2n \left(\frac{1}{\sqrt{2n}} \right)^2 = 1.$$

The two radial summands have opposite equal means and are independent. Proposition 5.1 computes

$$\int f_n^2 d\mu_n = \text{Var}_{\mu_n}(f_n) = \frac{4}{n+4} \rightarrow 0.$$

Consequently there is no positive dimension-free C_2 , and therefore the unrestricted general-isotropic question has a negative answer.

Scope and provenance

The counterexample does not resolve every possible “suitable symmetries” refinement. The 2026 paper makes one such refinement precise by imposing signed-permutation (H_n) invariance and states its all-degree L^2 version as Conjecture 1.4. This packet records only the full negative answer for arbitrary isotropic log-concave measures. The supporting authors cite the earlier paper and deliberately present the construction as showing that the symmetry assumption rules out pathological behavior, so this is an explicit later-literature resolution rather than a new result of the run.

References

- [1] I. Glazer and D. Mikulincer, *Anti-concentration of polynomials: dimension-free covariance bounds and decay of Fourier coefficients*, arXiv:2108.04268 (2021; revised 2022).
- [2] I. Glazer and D. Mikulincer, *Anti-concentration of polynomials: L^p balls and symmetric measures*, arXiv:2603.22664 (2026), Section 5.1 and Proposition 5.1.